

Radar Fields

An Extension of Radiance Fields to SAR

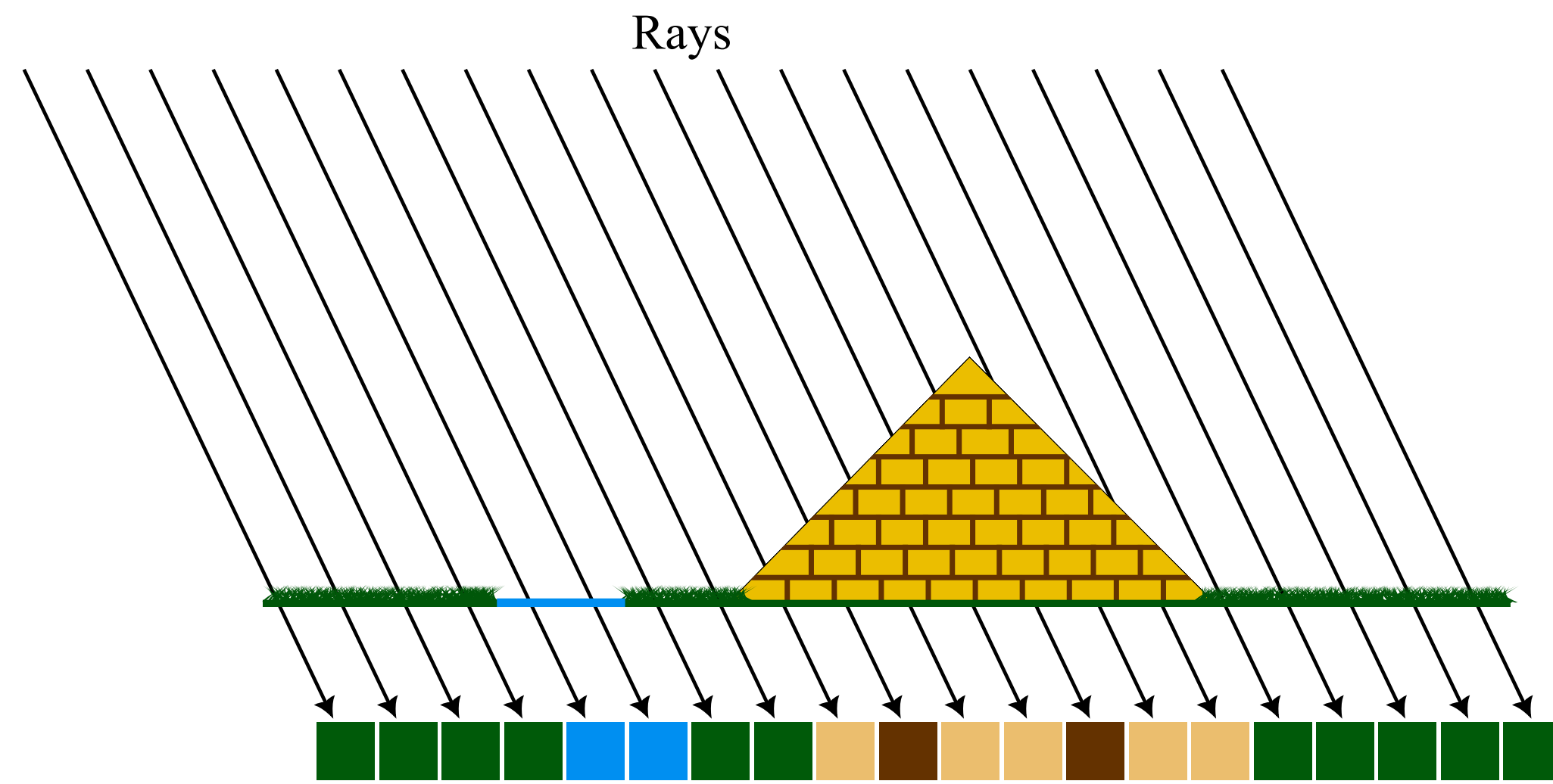
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Context

- Radiance fields (e.g. NeRFs [1]) were previously modified and successfully applied to remote sensing in [2].
- **This work:** Extends radiance fields concepts to Synthetic Aperture Radar (SAR) data.
- **Advantages of SAR:** See through clouds, smoke and during the night.
- **Example usecase:** Monitor volcanoes during eruption.

Radiance fields



Model definition

- For a position $\mathbf{x} \in \mathbb{R}^3$ and a view direction $\mathbf{v} \in \mathbb{R}^3$, the model is $\mathcal{F} : (\mathbf{x}, \mathbf{v}) \mapsto (\mathbf{c}, \sigma)$ with $\mathbf{c}$ the color and σ the density.

Opacity definition

$$\alpha_i = 1 - \exp(-\sigma_i(t_{i+1} - t_i))$$

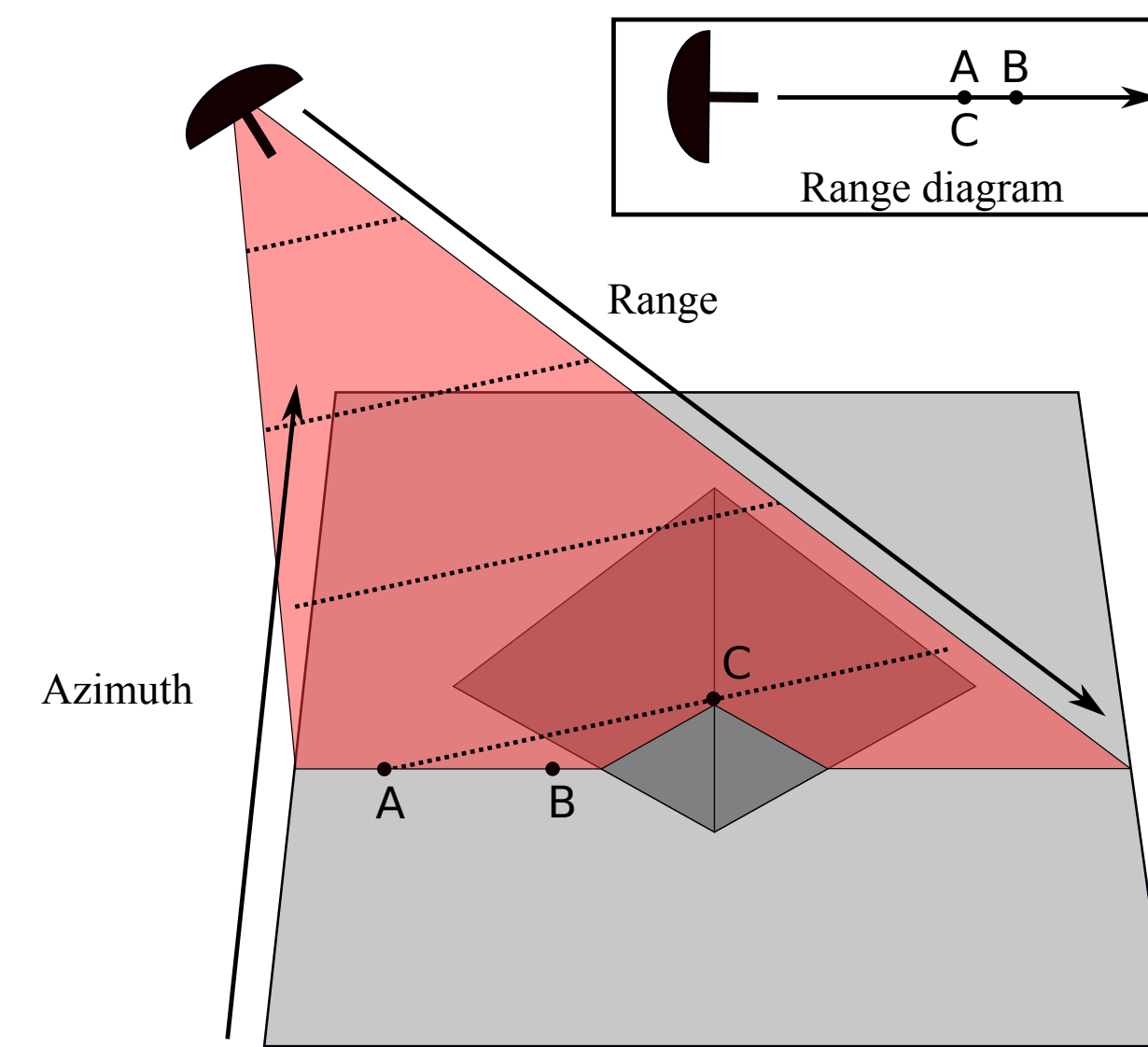
Transmittance definition

$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j).$$

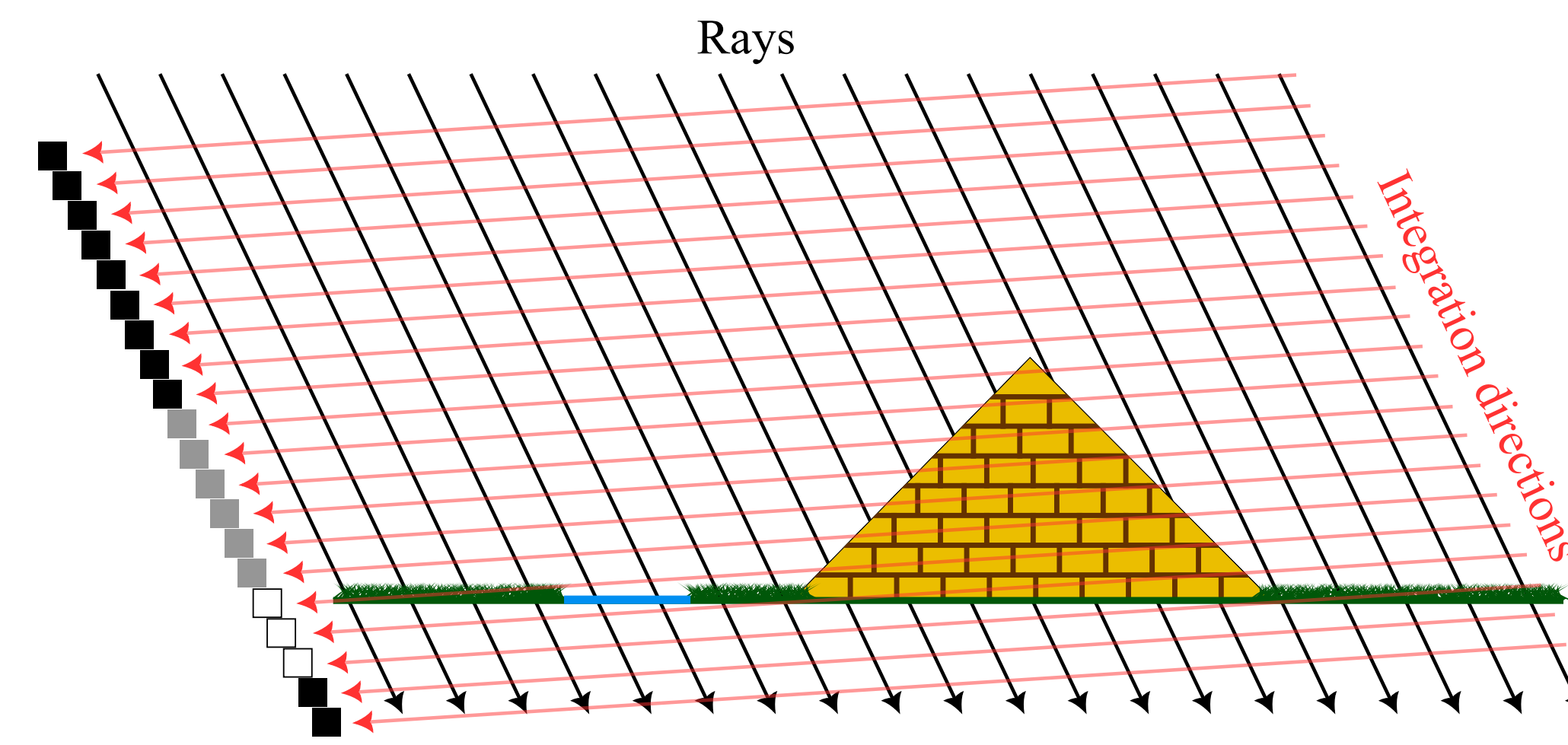
Color rendering along a ray $\mathbf{r}$

$$\mathbf{c}(\mathbf{r}) = \sum_{i=1}^N T_i \alpha_i \mathbf{c}_i.$$

Acquisition of a SAR image



Radar fields



Model definition

- For a position $\mathbf{x} \in \mathbb{R}^3$, the model is $\mathcal{F} : \mathbf{x} \in \mathbb{R}^3 \mapsto \sigma$ with σ the density.
- Reflected signal is defined as in [3, 4]: $s = -\langle \mathbf{v} | \nabla \mathcal{F}(\mathbf{x}) \rangle^\theta$ with $\mathbf{v} \in \mathbb{R}^3$ the view direction and $\theta \in \mathbb{R}^+$ the specularity parameter.

Opacity and transmittance

- Same as for radiance fields.

Radar rendering for a depth d_i

$$\begin{aligned} s(d_i) &= \sum_{\mathbf{r}_j} s_{\mathbf{r}_j}(d_i) \\ &= - \sum_{\mathbf{r}_j} T_{i,j} \sigma_{i,j} \langle \mathbf{v}_j | \mathbf{n}_{i,j} \rangle^\theta. \end{aligned}$$

Main differences

	Radiance field	Radar Field
Complexity	height $\times$ width $\times$ samples	azimuth $\times$ range $\times$ samples (on average)
Integration	Along the ray	Across the ray
Distance sampling	Variable (along a ray)	Fixed (defined by data)
Ray sampling	Fixed (defined by data)	Variable (in a given azimuth plane)
Noise	Noise-free	Speckle

Future challenges

From synthetic to real

- The current prototype only works with synthetic images.
- **Goal:** Extend the proof of concept to real images.

Multi-bounce

- Multiple reflections of the SAR signal can create virtual reflections.
- **Goal:** Model multi-bounce in the framework.

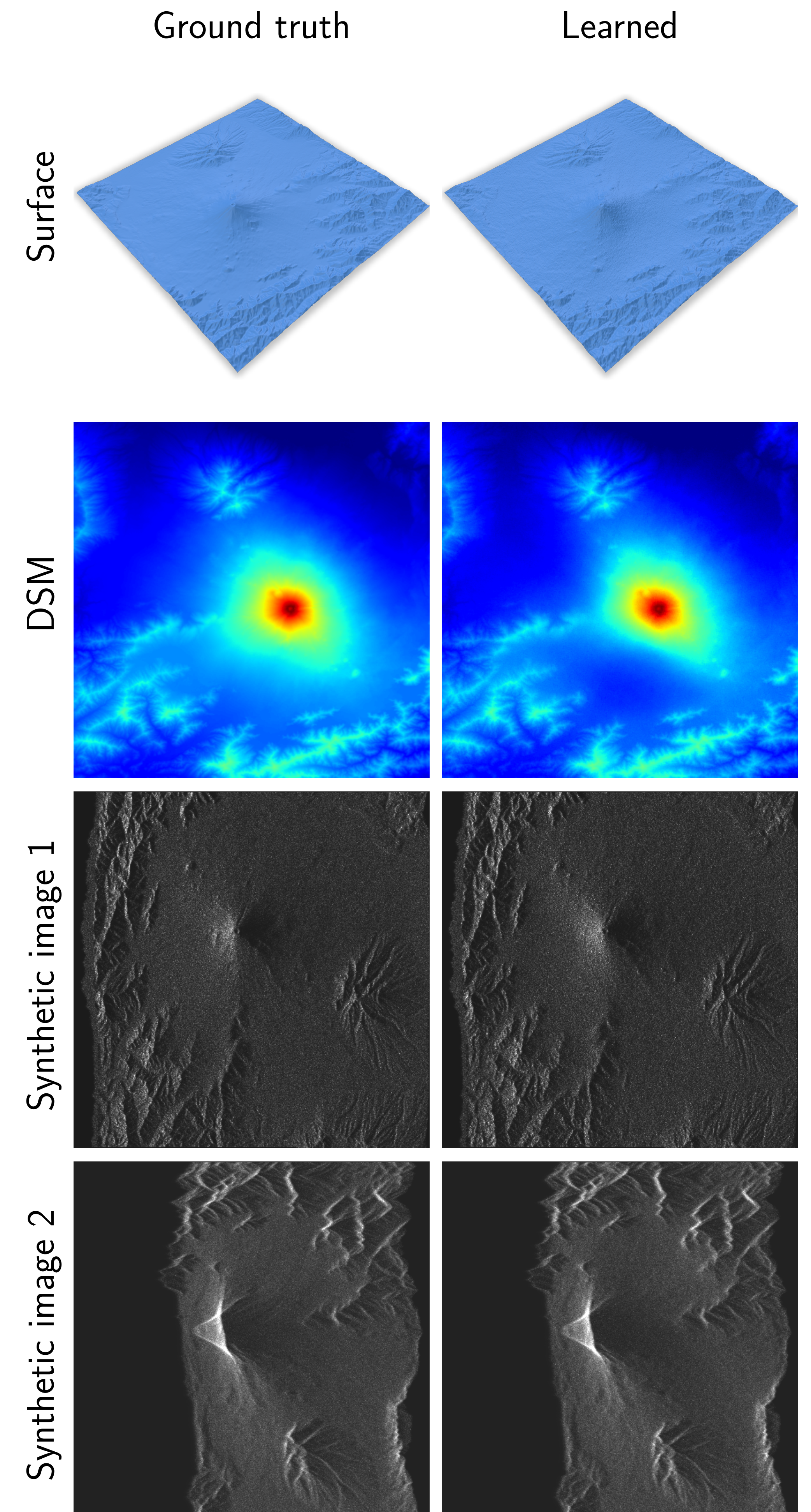
Phase and polarimetric information

- Phase information is frequently used to measure small terrain deformation.
- Polarimetric information is widely used to determine the scattering mechanism.
- **Goal:** Extend the model to take advantage of phase and polarimetry.

References

- [1] Mildenhall, Ben, et al. "NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis." *ECCV*, 2020.
- [2] Mari, Roger, et al. "Multi-Date Earth Observation NeRF: The Detail Is in the Shadows." *CVPRW, Earthvision*, 2023.
- [3] Balz, Timo, and Stilla, Uwe. "Hybrid GPU-based single-and double-bounce SAR simulation." *TGRS*, 2009.
- [4] "3D synthetic aperture radar simulation for interpreting complex urban reflection scenarios". *Thesis, Technische Universität München*, 2011.

Synthetic experiments



Output surface and rendered views using five input synthetic images.